## Question Paper

### Physics Single Correct (Maximum Marks: 80)

**Question No. 1**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Two bodies of masses $m_1$ and $m_2$ initially at rest at infinite distance apart move towards each other under gravitational force of attraction. Their relative velocity of approach when they are separated by a distance $r$ is ( $\mathrm{G}$ = universal gravitational constant.)

A. $\left[\frac{2G(m_1-m_2)}{r}\right]^{1/2}$

B. $\left[\frac{2G(m_1+m_2)}{r}\right]^{1/2}$

C. $\left[\frac{r}{2G(m_1m_2)}\right]^{1/2}$

D. $\left[\frac{r}{2G}\, m_1m_2\right]^{1/2}$

**Question No. 2**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

A wire of $20\ \Omega$ resistance is gradually stretched to double its original length. It is then cut into two equal parts. These parts are then connected in parallel across a 4.0 volt battery. The current drawn from the battery is:

A. 4 A

B. 2 A

C. 0.2 A

D. 2.2 A

JOIN - @IIT_JEE_NEET_BOOK

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Statement (A) Two artificial satellites revolving in the same circular orbit have same period of revolution.
Statement (B) The orbital velocity is inversely proportional to the square root of radius of the orbit.
Statement (C) The escape velocity of the body is independent of the altitude of the point of projection.

A. $A, B, C$ are true

B. A, B true $C$ false

C. $A, C$ true $B$ false

D. $B, C$ true $A$ false

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider the radiation emitted by large number of singly charged positive ions of a certain element. The sample emits fifteen types of spectral lines, one of which is same as the first line of Lyman series. What is the binding energy in the highest energy state of this configuration?

A. $13.6$ eV

B. $54.4$ eV

C. $10.2$ eV

D. $1.6$ eV

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A. Wavelength of microwaves is greater than that of ultraviolet rays.
B. The wavelength of infrared rays is lesser than that of ultraviolet rays.
C. The wavelength of microwaves is lesser than that of infrared rays.
D. Gamma-ray has the shortest wavelength in the electromagnetic spectrum.
What should be the correct statement from the given statements ?

A. A and B are true

B. B and C are true

C. C and D are true

D. A and D are true

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A charge $Q_1$ exerts force on a second charge $Q_2$. If a 3rd charge $Q_3$ is brought near, the force of $Q_1$ exerted on $Q_2$,

A. Will increase

B. Will decrease

C. Will remain unchanged

D. Will increase if $Q_3$ is of the same sign as $Q_1$ and will decrease if $Q_3$ is of opposite sign

JOIN - @IIT_JEE_NEET_BOOK

**Question No. 7**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

When an object is shot from the bottom of a long smooth inclined plane kept at an angle $60^\circ$ with horizontal, it can travel a distance $x_1$ along the plane. But, when the inclination is decreased to $30^\circ$ and the same object is shot with the same velocity, it can travel $x_2$ distance. Then $x_1 : x_2$ will be.....

A. $1 : \sqrt{2}$

B. $\sqrt{2} : 1$

C. $1 : \sqrt{3}$

D. $1 : 2\sqrt{3}$

**Question No. 8**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Which of the following are true regarding image formation with the help of mirrors?
I. A concave mirror can give a diminished virtual image.
II. A concave mirror can give a real image.
III. A convex mirror can give a virtual image.
IV. A convex mirror cannot give a real image.
Select the correct answer from the options given below.

A. Both I and III

B. Both I and IV

C. I, III and IV

D. II, III and IV

JOIN - @IIT_JEE_NEET_BOOK

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A particle is performing simple harmonic motion along $x-$ axis with amplitude $4$ cm and time period $1.2$ sec. The minimum time taken by the particle to move from $x=+2$ cm to $x=+4$ cm and back again is given by:

A. $0.6$ sec

B. $0.4$ sec

C. $0.3$ sec

D. $0.2$ sec

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If one mole of the polyatomic gas is having two vibrational modes and $\beta$ is the ratio of molar specific heats for polyatomic gas $\left(\beta=\frac{C_P}{C_v}\right)$ then the value of $\beta$ is :

A. $1.02$

B. $1.2$

C. $1.25$

D. $1.35$

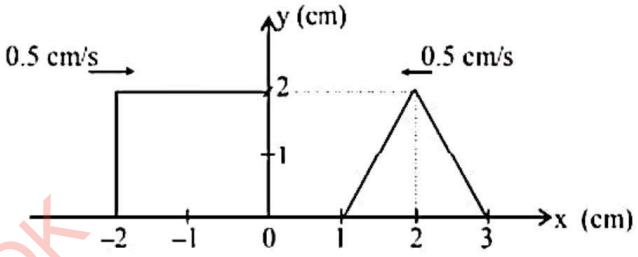
JOIN - @IIT_JEE_NEET_BOOK

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Figure shows a rectangular pulse and a triangular pulse approaching each other along x-axis. The pulse speed is $0.5$ cm/s. What is the resultant displacement of medium particles due to superposition of waves at $x = 0.5$ cm and $t = 2$sec.




A. $3.5$ cm

B. $2.5$ cm

C. $4$ cm

D. $3$ cm

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match **List I** with **List II**.

| List I | | List II | |
|---|---|---|---|
| (a) | Isothermal | (i) | Pressure constant |
| (b) | Isochoric | (ii) | Temperature constant |
| (c) | Adiabatic | (iii) | Volume constant |
| (d) | Isobaric | (iv) | Heat content is constant |

Choose the correct answer from the options given below:

A. $(a) \rightarrow (ii), (b) \rightarrow (iii), (c) \rightarrow (iv), (d) \rightarrow (i)$

B. $(a) \rightarrow (iii), (b) \rightarrow (ii), (c) \rightarrow (i), (d) \rightarrow (iv)$

C. $(a) \rightarrow (i), (b) \rightarrow (iii), (c) \rightarrow (ii), (d) \rightarrow (iv)$

D. $(a) \rightarrow (ii), (b) \rightarrow (iv), (c) \rightarrow (iii), (d) \rightarrow (i)$

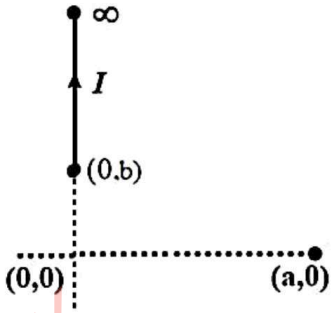
JOIN - @IIT_JEE_NEET_BOOK

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An infinitely long wire carrying current $i$ is along $Y$-axis such that its one end is at point $(0, b)$ while the wire extends upto $\infty$. The magnitude of magnetic field strength at point $P(a, 0)$ is




A. $\frac{\mu_0 i}{4\pi a}\left(1+\frac{b}{\sqrt{a^2+b^2}}\right)$

B. $\frac{\mu_0 i}{4\pi a}\left(1-\frac{b}{\sqrt{a^2+b^2}}\right)$

C. $\frac{\mu_0 i}{4\pi a}\left(1-\frac{a}{\sqrt{a^2+b^2}}\right)$

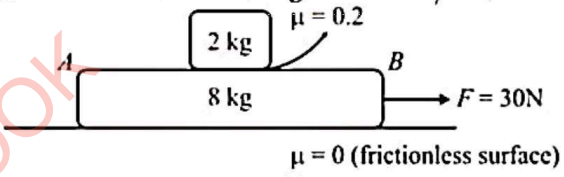
D. $\frac{\mu_0 i}{4\pi a}\left(1+\frac{a}{\sqrt{a^2+b^2}}\right)$

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two blocks are placed as shown in the figure. The upper block is placed at 2 m from the end $A$. Find (approximate) time taken by 2 kg box to fell from A.
(Take the value of $g$ as $10\mathrm{m/s^2}$)




A. $\sqrt{3}s$

B. 2 s

JOIN - @IIT_JEE_NEET_BOOK

C. 3 s

D. 1.2 s

**Question No. 15**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Match List I with List II

| List I | List II |
|---|---|
| (A) Angular Momentum (L) | $(I)\left[ML^2T^{-2}\theta^{-1}\right]$ |
| (B) Entropy (S) | $(II)\left[ML^2T^{-1}\right]$ |
| (C) Surface Tension (T) | $(III)\left[ML^2T^{-2}A^{-1}\right]$ |
| (D) Magnetic Flux ($\Phi$) | $(IV)\left[MT^{-2}\right]$ |

A. (A - IV, B - II, C - I, D - III)

B. (A - III, B - IV, C - II, D - I)

C. (A - II, B - I, C - IV, D - III)

D. (A - II, B - IV, C - III, D - I)

JOIN - @IIT_JEE_NEET_BOOK

**Question No. 16**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Assertion: For a point light source, frequency of light emitted is numerically equal to number of photons emitted per second.
Reason: According to wave theory frequency of light wave is equivalent to number of cycles completed per second and wave-particle duality explains the dual nature of electromagnetic waves as a wave or as a corpuscle.

A. If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but Reason is not explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If Assertion is false but Reason is true.

**Question No. 17**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

A car is moving along a straight road with uniform acceleration. It passes through two points $P$ and $Q$ separated by a distance with velocities $30\ \text{kmh}^{-1}$ and $40\ \text{kmh}^{-1}$ respectively. The velocity of car midway between $P$ and $Q$ is?

A. $33.3\ \text{kmh}^{-1}$

B. $1\ \text{kmh}^{-1}$

C. $25\sqrt{3}\ \text{kmh}^{-1}$

D. $35.35\ \text{kmh}^{-1}$

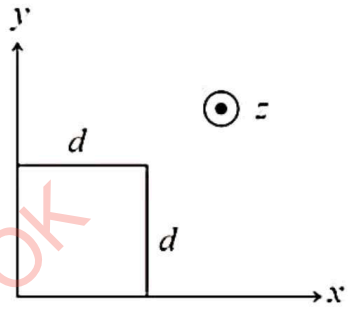
JOIN - @IIT_JEE_NEET_BOOK

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The magnetic field in a region is given by $\vec{B} = B_0\left(\frac{x}{a}\right)\hat{k}$. A square loop of side d is placed with its edges along the $x$ and $y$ axes. The loop is moved with a constant velocity $\vec{v} = v_0\hat{i}$. The emf induced in the loop is :




A. $\frac{B_0 v_0^2\, d}{2a}$

B. $\frac{B_0 v_0 d}{2a}$

C. $\frac{B_0 v_0 d^2}{a}$

D. $\frac{B_0 v_0 d^2}{2a}$

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

On an X temperature scale, water freezes at $-125.0°\text{X}$ and boils at $375.0°\text{X}$. On a Y temperature scale, water freezes at $-70.0°\text{Y}$ and boils at $-30.0°\text{Y}$. The value of temperature on X-scale equal to the temperature of $50.0°\text{Y}$ on Y-scale is :

A. $455.0°\text{X}$

B. $-125.0°\text{X}$

C. $1375.0°\text{X}$

D. $1500.0°\text{X}$

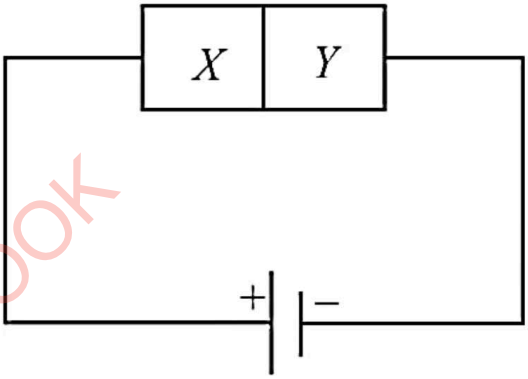
JOIN - @IIT_JEE_NEET_BOOK

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A semiconductor $X$ is made by doping a germanium crystal with arsenic $(Z = 33)$. A second semiconductor $Y$ is made by doping germanium with indium $(Z = 49)$. The two are joined end to end and connected to a battery as shown in the figure. Which of the following statements is correct?




A. $X$ is $p$-type, $Y$ is $n$-type and the junction is forward biased.

B. $X$ is $n$-type, $Y$ is $p$-type and the junction is forward biased.

C. $X$ is $p$-type, $Y$ is $n$-type and the junction is reverse biased.

D. $X$ is $n$-type, $Y$ is $p$-type and the junction is reverse biased.

## Physics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

Two bodies of mass $4$ kg and $5$ kg are moving along east and north directions with velocities $5$ m/s and $3$ m/s respectively. Magnitude of the velocity of centre of mass of the system is Vcm, find 18 Vcm.

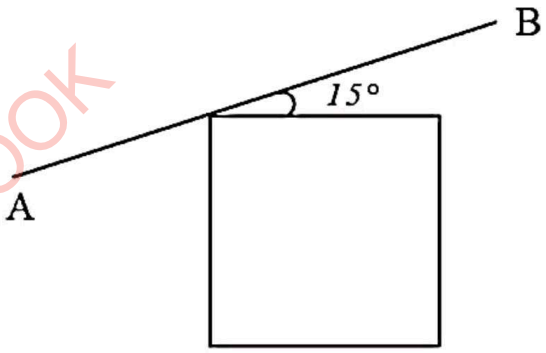
JOIN - @IIT_JEE_NEET_BOOK

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

A square plate of mass M and edge L is shown in figure. Moment of inertia of the plate about the axis in the plane of plate and passing through one of its vertex making an angle $15^\circ$ with its side as shown in figure is $xML^2/24$, find $x$.




Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

The closed and open organ pipes have same length. When they are vibrating simultaneously in first overtone, produce three beats. The length of open pipe is made $\frac{1}{3}$ rd and closed pipe is made three times the original, when they are vibrating in fundamental mode, the number of beats produced will be....

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

A point source S is placed at a distance of 15 cm from a converging lens of focal length 10 cm on its principal axis. The diverging mirror of focal length 12 cm is placed at a certain distance from lens such that a real image is formed on the source itself. Find the minimum distance between lens and mirror.

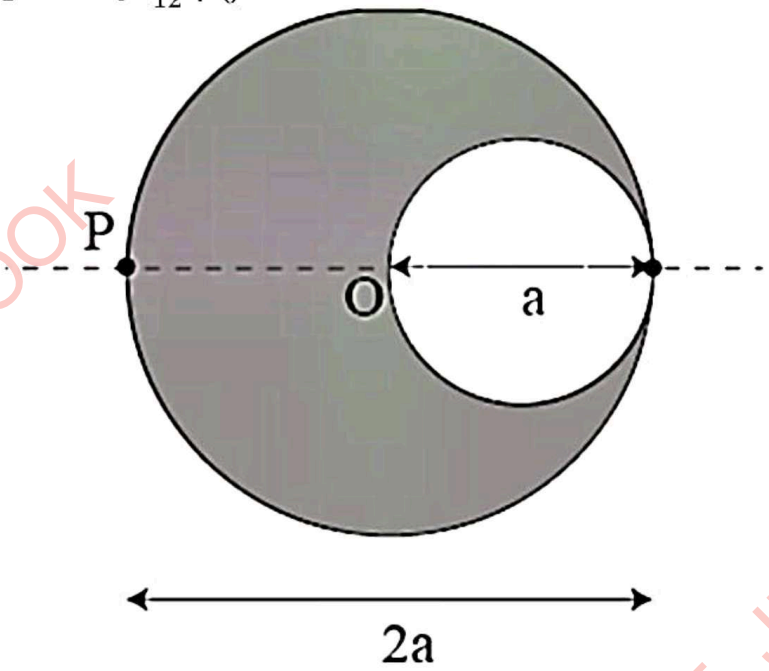
JOIN - @IIT_JEE_NEET_BOOK

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

A cylindrical cavity of diameter a exists inside a cylinder of diameter $2a$ as shown in the figure. Both the cylinder and the cavity are infinity long. A uniform current density $J$ flows along the length. If the magnitude of the magnetic field at the point $P$ is given by $\frac{N}{12}\mu_0 aJ$, then the value of N is




## Chemistry Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following lanthanoid ions is diamagnetic?

(At nos. $\mathrm{Ce} = 58, \mathrm{Sm} = 62, \mathrm{Eu} = 63, \mathrm{Yb} = 70$):

A. $\mathrm{Sm}^{2+}$

B. $\mathrm{Eu}^{2+}$

C. $\mathrm{Yb}^{2+}$

D. $\mathrm{Ce}^{2+}$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Incorrect order of radius is.....

A. $Sr^{2+} < Rb^{+} < Br^{-} < Se^{2-}$

B. $Nb^{5+} < Zr^{4+} < Y^{3+}$

C. $Co > Co^{2+} > Co^{3+} > Co^{4+}$

D. $Ba^{2+} > Cs^{+} > Se^{2-} > As^{3-}$

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : In $TlI_3$, isomorphous to $CsI_3$, the metal is present in $+1$ oxidation state.

Reason R : $Tl$ metal has fourteen f electrons in its electronic configuration.

In the light of the above statements, choose the most appropriate answer from the options given below:

A. A is correct but R is not correct

B. Both A and R are correct and R is the correct explanation of A

C. A is not correct but R is correct

D. Both A and R are correct but R is NOT the correct explanation of A

JOIN - @IIT_JEE_NEET_BOOK

**Question No. 4**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

The dissociation constant of two acids $HA_1$ and $HA_2$ are $3.14 \times 10^{-4}$ and $1.96 \times 10^{-5}$ respectively. The relative strength of the acids will be approximately.

A. $1 : 4$

B. $4 : 1$

C. $1 : 16$

D. $16 : 1$

**Question No. 5**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

For which of the following, Kjeldahl's method is not used for the estimation of nitrogen?

I. Aniline

II. Azobenzene

III. Nitrobenzene

IV. Pyridine

A. II, III, IV

B. II, III only

C. III, IV only

D. I, III, IV

JOIN - @IIT_JEE_NEET_BOOK

**Question No. 6**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Given that : $E^{\circ}_{O_2/H_2O} = +1.23V$ ,
$E^{\circ}_{S_2O_8^{2-}/SO_4^{2-}} = +2.05V$
$E^{\circ}_{Br_2/Br^-} = +1.09V$
$E^{\circ}_{Au^{3+}/Au} = +1.4V$
The strongest oxidizing agent is

A. $O_2$

B. $Br_2$

C. $S_2O_8{}^{2-}$

D. $Au^{3+}$

**Question No. 7**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

The weight in grams of a non-volatile solute (mol. wt. 60) to be dissolved in 90 g of water to produce a relative lowering of vapour pressure of 0.02 is---

A. 4

B. 8

C. 6

D. 10

JOIN - @IIT_JEE_NEET_BOOK

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match List-I with List-II. Ionization Energy is mentioned in List - II.

| List-I<br>Electronic configuration of elements | List-II<br>$\Delta_i H$ in kJ $mol^{-1}$ |
|---|---|
| (a) $1s^2 2\ s^2$ | (p) 801 |
| (b) $1s^2 2s^2 2p^4$ | (q) 899 |
| (c) $1\ s^2 2\ s^2 2p^3$ | (r) 1314 |
| (d) $1\ s^2 2\ s^2 2p^1$ | (s) 1402 |

Choose the most appropriate answer from the options given below:

A. (a) $\rightarrow$ (q), (b) $\rightarrow$ (r), (c) $\rightarrow$ (s), (d) $\rightarrow$ (p)

B. (a) $\rightarrow$ (p), (b) $\rightarrow$ (r), (c) $\rightarrow$ (s), (d) $\rightarrow$ (q)

C. (a) $\rightarrow$ (p), (b) $\rightarrow$ (s), (c) $\rightarrow$ (r), (d) $\rightarrow$ (q)

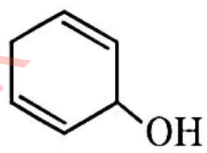
D. (a) $\rightarrow$ (s),(b) $\rightarrow$ (p), (c) $\rightarrow$ (q), (d) $\rightarrow$ (r)

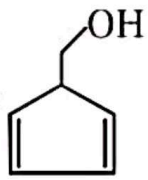
Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

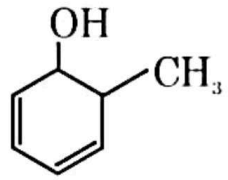
Which alcohols on reaction with concentrated $H_2SO_4$ at high temperature will give product, which give +ve test with $Br_2$ solution :

A. 


B. 


JOIN - @IIT_JEE_NEET_BOOK

C. $Me_3C - CH_2 - CH_2OH$

D. 

**Question No. 10**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Arrange the following anilines in decreasing order of basicity

1. $C_6H_5NH_2$
2. o $- CH_3C_6H_4NH_2$
3. m $- CH_3C_6H_4NH_2$
4. p $- CH_3C_6H_4NH_2$

A. 4 > 1 > 2 > 3

B. 4 > 3 > 1 > 2

C. 1 > 2 > 3 > 4

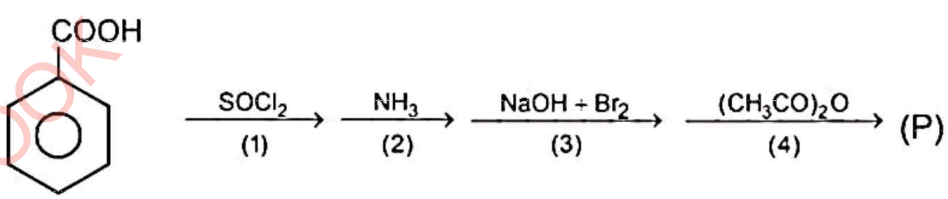
D. 4 > 3 > 2 > 1

**Question No. 11**

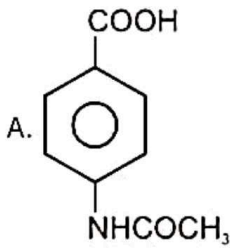
**Only One Option Correct Type**

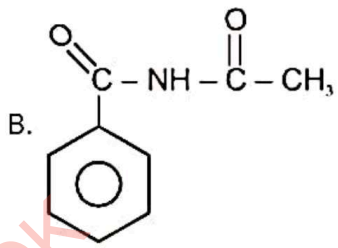
Each question has multiple options out of which ONLY ONE is correct.

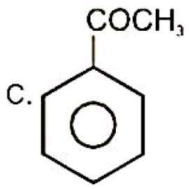
The structure of product $(P)$ is :

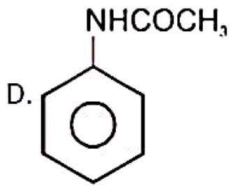



JOIN - @IIT_JEE_NEET_BOOK

A. 


B. 


C. 


D. 


**Question No. 12**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

The rate of reaction is doubled for every $10^{\circ}C$ rise in temperature. The increase in rate as result of an increase in temperature from $10^{\circ}C$ to $100^{\circ}C$ is how many times of the original rate?

A. 112

B. 512

C. 400

D. 256

JOIN - @IIT_JEE_NEET_BOOK

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following statements is/are correct when a mixture of $HgCl_2$ and $K_2Cr_2O_7$ is gently warmed with concentrated $H_2SO_4$?

a. A deep red vapour is evolved.

b. The vapour when passed into NaOH solution gives a yellow solution of $Na_2CrO_4$.

c. The reaction indicates presence of $Cl^-$ ion in $HgCl_2$.

d. Chromyl chloride is formed.

A. a, b, and d

B. a, b, c and d

C. b and c

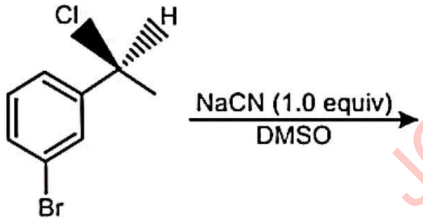
D. None of these.

Question No. 14

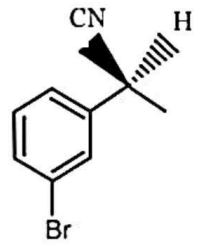
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

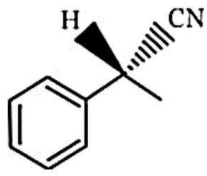
Predict the major product of the following reaction;



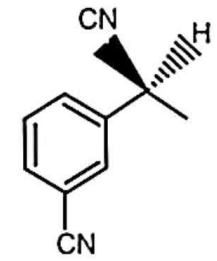

A.

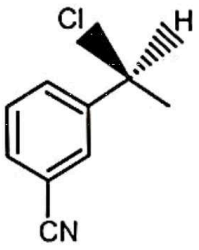



B.




JOIN - @IIT_JEE_NEET_BOOK

C. 


D. 


**Question No. 15**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Match the following.

| | Column I | | Column II |
|---|---|---|---|
| A. | Acetaldehyde, vinyl alcohol | 1. | Enantiomers |
| B. | Eclipsed and staggered ethane | 2. | Tautomers |
| C. | (+)2-butanol, (-)2-butanol | 3. | Chain isomers |
| D. | Methyl-n-propyl-amine and diethylamine | 4. | Conformational isomers |
| | | 5. | Metamers |

A, B, C, D match with

A. 1 4 3 5

B. 2 4 1 5

C. 5 1 4 2

D. 5 1 3 2

JOIN - @IIT_JEE_NEET_BOOK

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The correct order of boiling point is

A. $T_2 < D_2 < H_2$

B. n-pentane < neo-pentane

C. $Xe < Ar < He$

D. m-nitrophenol > o-nitrophenol

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The solutions of two compounds listed in Column-I when taken in molecular proportions give coordination compounds on evaporation. Match these compounds with their relative properties which is/are listed in Column-II.

| Column-I | Column-II |
|---|---|
| A) $Fe(CN)_2 + 4\,KCN$ | p) Coordination number of central metal ion in six |
| B) $CuSO_4 + 4\,NH_3$ | q) Coordination number of central metal ion is four |
| C) $3\,NaF + AlF_3$ | r) Coordination number of central metal ion is two |
| D) $AgCN + KCN$ | s) Oxidation state of central metal ion is +2 |
| | t) Overall charge on complex ion is +2 |

?

A. A-p,s ; B-q,s,t; C-p; D-r

B. A-q ; B-q,s,t; C-p; D-r

C. A-s ; B-q,s,t; C-p; D-r

D. A-p; B-q,s,t; C-p; D-r

**Question No. 18**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Considering the elements B, C, N, F and Si, the correct order of their non-metallic character is--

A. B > C > Si > N > F

B. Si > C > B > N > F

C. F > N > C > B > Si

D. F > N > C > Si > B

**Question No. 19**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Assertion : Tertiary alcohols produce turbidity immediately with Lucas reagent.
Reason : Lucas reagent is a $1:1$ mixture of conc. $HNO_3$ and anhydrous $ZnCl_2$.
The correct option among the following is---

A. (A) is true, (R) is true and (R) is the correct explanation for (A)

B. (A) is true, (R) is true but (R) is not the correct explanation for (A)

C. (A) is true but (R) is false

D. (A) is false but (R) is true

JOIN - @IIT_JEE_NEET_BOOK

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The unstable green colored $K_2MnO_4$ is converted to purple color. Then the correct statements are
a. It is a disproportionation reaction.
b. The product formed is $KMnO_4$.
c. Nature of the overall solution becomes alkaline.
d. It produces black precipitates of $MnO_2$.

A. a and b but not c and d

B. a, b, c and d are correct

C. a and d but not b and c

D. b, c and d but not a

## Chemistry Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

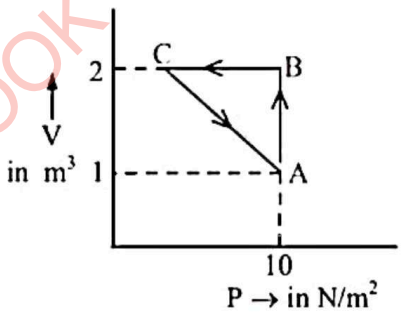
The number of Faradays required to completely deposit magnesium from 1 L of $0.1M$ $MgCl_2$ aq. solution is K, find 20K

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

An ideal gas is taken through the cycle $A \longrightarrow B \longrightarrow C \longrightarrow A$ as shown in the figure. If net heat supplied to the gas in the cycle is 5 J, the work done on the gas in the process $C \rightarrow A$ is _____ J.




JOIN - @IIT_JEE_NEET_BOOK

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

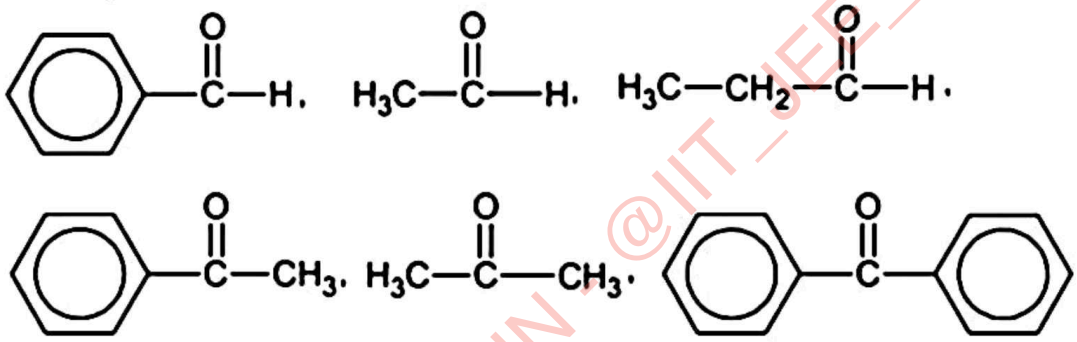
In a neutral or faintly alkaline solution, 8 moles of permanganate anion quantitatively oxidize thiosulphate anions to produce X moles of a sulphur containing product. The magnitude of X is:

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

Among the following the number of compounds which react with Fehling solution is:



Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

For the $1^{st}$ order reaction: $A(g) \rightarrow 2B(g) + C\ (s\ )$; the $t_{\frac{1}{2}} = 24\ min$. The reaction is carried out taking certain mass of A enclosed in a vessel in which it exerts a pressure of 400 mm Hg. The pressure of the reaction mixture in mm Hg after expiry of 48 min will be

JOIN - @IIT_JEE_NEET_BOOK

## Mathematics Single Correct (Maximum Marks: 80)

**Question No. 1**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

If $y(x)$ is a solution of $\frac{dy}{dx} - \frac{xy}{1+x} = \frac{1}{1+x}$ and $y(0) = -1$, then the value of $y(2)$ is

A. $-\frac{1}{2}$

B. $-\frac{1}{3}$

C. $-\frac{1}{4}$

D. $-\frac{1}{5}$

**Question No. 2**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

The number of ways of arranging the letters AAAAA, BBB, CCC, D, EE and F in a row, if the letters B are separated from one another, is equal to

A. $\frac{13!}{5!3!3!2!}$

B. $\frac{14!}{3!3!2!}$

C. $\frac{15!}{(3!)^2 2!5!}$

D. ${}^{13}C_3 \times \frac{12!}{5!3!2!}$

JOIN - @IIT_JEE_NEET_BOOK

**Question No. 3**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Let $V_1 =$ variance of $\{13,16,19.\ldots\ldots\ldots 103\}$ and $V_2 =$ variance of $\{20,26,32.\ldots\ldots\ldots 200\}$. Then $V_1 : V_2$ is

A. $1:2$

B. $1:1$

C. $4:9$

D. $1:4$

**Question No. 4**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

If $PQ$ be a double ordinate of the parabola, $y^2 = -4x$, where $P$ lies in the second quadrant. If $R$ divides $PQ$ in the ratio $2:1$, then the locus of $R$ is:

A. $3y^2 = -2x$

B. $9y^2 = 4x$

C. $9y^2 = -4x$

D. $3y^2 = 2x$

JOIN - @IIT_JEE_NEET_BOOK

**Question No. 5**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

A pair of students is selected at random from a probability class. The probability that the pair selected will consist of one male and one female student is $\frac{10}{19}$. Find the maximum number of students the class can contain.

A. 15

B. 20

C. 28

D. 24

**Question No. 6**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

$A,\ B,\ C$ are three matrices of the same order such that any two are symmetric and the $3^{\text{rd}}$ one is Skew-Symmetric. If $X = ABC + CBA$ and $Y = ABC - CBA$, then $(XY)^T$ is

A. Symmetric

B. Skew-Symmetric

C. $I - XY$

D. $-YX$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For a function $f(x) = \frac{2(x^2+1)}{[x]}$ (where [.] denotes the greatest integer function), if $1 \leq x < 4$, then

A. the range of $f$ is $\left[4, \frac{17}{2}\right)$

B. $f$ is bijective function

C. the maximum value of $f(x)$ is $\frac{34}{3}$

D. the minimum value of $f(x)$ is $4$

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the shortest distance between the lines $\mathbf{r} = (3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 2\hat{\mathbf{k}}) + t(-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}})$ and $\mathbf{r} = (\hat{\mathbf{i}} - 7\hat{\mathbf{j}} - 2\hat{\mathbf{k}}) + s(\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}})$ is equivalent to projection of $\mathbf{P} = -2\hat{\mathbf{i}} + 11\hat{\mathbf{j}}$ on $\mathbf{Q}$ then a possible vector $\mathbf{Q}$ is

A. $\hat{i} + 5\hat{j} - 3\hat{k}$

B. $5\hat{i} - \hat{j} + 3\hat{k}$

C. $-1\hat{\mathrm{i}} + 3\hat{\mathrm{j}} + 5\hat{\mathrm{k}}$

D. $3\hat{i} + 5\hat{j} - \hat{k}$

JOIN - @IIT_JEE_NEET_BOOK

**Question No. 9**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

The area bounded by y $= f(\mathrm{x})$, x-axis, y-axis and the line y $= 1$ whose $f(\mathrm{x}) = 1 + \frac{1}{x}\int_1^x f(t)dt, x > 0$ is

A. $2(\mathrm{e}+1)$

B. $2\left(1-\frac{1}{e}\right)$

C. $2(\mathrm{e}-1)$

D. $\left(1-\frac{1}{e}\right)$

**Question No. 10**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Given three non-zero, non-coplanar vectors $\vec{a}, \vec{b}\&\vec{c}$ and no two of which are collinear. $\overrightarrow{r_1} = p\vec{a} + q\vec{b} + \vec{c}$ and $\overrightarrow{r_2} = \vec{a} + p\vec{b} + q\vec{c}$, if the vectors $\overrightarrow{r_1} + 2\overrightarrow{r_2}$ and $2\overrightarrow{r_1} + \overrightarrow{r_2}$ are collinear then value of $(p+q)$ is

A. 3

B. 4

C. 3.5

D. 2

JOIN - @IIT_JEE_NEET_BOOK

**Question No. 11**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

A perpendicular is drawn from the point $P(2,\ 4,\ -1)$ to the line $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$ . The equation of the perpendicular from P to the given line is

A. $\frac{x-2}{6} = \frac{y-4}{3} = \frac{z+1}{2}$

B. $\frac{x+2}{6} = \frac{y-4}{3} = \frac{z+1}{2}$

C. $\frac{x+2}{-6} = \frac{y-4}{3} = \frac{z+1}{2}$

D. $\frac{x+2}{6} = \frac{y+4}{3} = \frac{z+1}{2}$

**Question No. 12**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

If $f(\mathrm{x})$ is a positive differentiable function satisfying $f(\mathrm{x}+\mathrm{y}+\mathrm{z}) = f(\mathrm{x}) \cdot f(\mathrm{y}) \cdot f(\mathrm{z}) \forall \mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathrm{R}$ and $f(2) = 4, f'(0) = 2$, then $f'(2)$ equals

A. 6

B. 4

C. 8

D. 12

JOIN - @IIT_JEE_NEET_BOOK

**Question No. 13**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Let $f(b)$ be the minimum value of the expression $y = x^2 - 2x + \left(b^3 - 3b^2 + 4\right)\ \forall x \in R$. Then, the maximum value of $f(b)$ as $b$ varies from $0$ to $4$ is

A. $20$

B. $19$

C. $63$

D. $64$

**Question No. 14**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Let $f$ be a function from the set of positive integers to the set of real numbers. If: $\mathbb{N} \to \mathbb{R}$ such that
(i) $f(1) = 1$
(ii) $f(1) + 2f(2) + 3f(3) + \cdots + nf(n) = n(n+1)f(n)$.
Find $1/f(1997)$.

A. $4004$

B. $3994$

C. $3884$

D. $4114$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$\vec{b}$ and $\vec{c}$ are non collinear vectors and $(\vec{c}\cdot\vec{c})\vec{a}=\vec{c}$.

If
$(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}+(\vec{a}\cdot\vec{b})\vec{b}=(4-2\beta-\sin\alpha)\vec{b}+\left(\beta^2-1\right)\vec{c}$
then $\sin(\alpha+\beta)=$

A. $0$

B. $1$

C. $\sin 1$

D. $\cos 1$

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If sum of middle terms in the expansion of $\left(2a-\frac{a^2}{4}\right)^9$ is $S$, then the value of $S$ is:

A. $\left(\frac{63}{32}\right)a^{14}(8+a)$

B. $\left(\frac{63}{32}\right)a^{13}(8+a)$

C. $\left(\frac{63}{32}\right)a^{14}(8-a)$

D. $\left(\frac{63}{32}\right)a^{13}(8-a)$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of the definite integral $\int_{-\infty}^{a} \frac{\left(\sin^{-1}e^{x}+\sec^{-1}e^{-x}\right)dx}{\left(\tan^{-1}e^{a}+\tan^{-1}e^{x}\right)\left(e^{x}+e^{-x}\right)}\,(a \in R)$ is :

A. $\ln 3$

B. dependent on a

C. $\frac{\pi}{2}\,\ell n2$

D. $\frac{\pi}{2}\,\ell n\left(2\tan^{-1}e^{a}\right)$

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $B=\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$ and $A$ be a $2\times 2$ matrix satisfying $\left(A^{T}\right)^{-1}=A$.

If $X=ABA^{T}$, then $A^{T}X^{2021}A=$

A. $\begin{bmatrix}1 & 2^{2021}\\ 0 & 1\end{bmatrix}$

B. $\begin{bmatrix}1 & 2021\\ 0 & 1\end{bmatrix}$

C. $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

D. $\begin{bmatrix}1 & 4042\\ 0 & 1\end{bmatrix}$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For a complex number $Z$, if the argument of $(Z-a)\left(\overline{Z}-b\right)$ is $\frac{\pi}{4}$ or $\frac{-3\pi}{4}$ (where $a, b$ are two real numbers), then the value of $ab$ such that the locus of $Z$ represents a circle with centre $\frac{3}{2}+\frac{i}{2}$ is (where, $i^2=-1$)

A. $1$

B. $2$

C. $3$

D. $4$

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For $a, b>0$, let $5a-b, 2a+b, a+2b$ be in A.P. and $(b+1)^2, ab+1, (a-1)^2$ are in G.P., then compute $\left(a^{-1}+b^{-1}\right)$

A. $7$

B. $6$

C. $5$

D. $2$

JOIN - @IIT_JEE_NEET_BOOK

## Mathematics Numerical (Maximum Marks: 20)

**Question No. 1**

Numerical Type

The answer has to be filled into the input box provided below.

If $I_k = \int_{-2k\pi}^{2k\pi} |\sin x|[\sin x]dx, \forall k \in N$, where [-] denotes the greatest integer function, then $\sum_{k=1}^{10} I_k$ is equal to $P$, find $|P|$.

**Question No. 2**

Numerical Type

The answer has to be filled into the input box provided below.

The number of ways in which $8$ non-identical apples can be distributed among $3$ boys such that every boy should get atleast $1$ apple and atmost $4$ apples is $k \cdot P_3^7$, where $k$ has the value equal to

**Question No. 3**

Numerical Type

The answer has to be filled into the input box provided below.

If $A\left(n, n^2\right)$ ( where, $n \in N$) is any point in the interior of the quadrilateral formed by the lines $x = 0$, $y = 0$, $3x + y - 4 = 0$ and $4x + y - 21 = 0$, then the possible number of positions of the point $A$ is

**Question No. 4**

Numerical Type

The answer has to be filled into the input box provided below.

If $\alpha$ and $\beta$ are the roots of the equation $x^2 - 2x + 3 = 0$, then the sum of roots of the equation having roots as $\alpha^3 - 3\alpha^2 + 5\alpha - 2$ and $\beta^3 - \beta^2 + \beta + 5$ is

**Question No. 5**

Numerical Type

The answer has to be filled into the input box provided below.

$2025^{2025}$ is divided by 18 , find the remainder.

JOIN_@IIT_JEE_NEET_BOOK